

LUND UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written examination (resit)

The use of electronic devices and additional material (books, notes, etc.) is not permitted during the exam. Justify clearly your answers and please write your anonymous code at the top of each sheet.

1. The *Liouville function* λ is defined by $\lambda(1) = 1$ and $\lambda(n) = (-1)^{a_1+\cdots+a_k}$ if $n > 1$ and $n = p_1^{a_1} \cdots p_k^{a_k}$ is the prime factorisation of n.
   (a) Prove that λ is a multiplicative function.
   (b) Given a positive integer n, prove that
   $$\sum_{d|n} \lambda(d) = \begin{cases} 1 & \text{if } n = m^2 \text{ for some integer } m, \\ 0 & \text{otherwise.} \end{cases}$$
2. (a) Show that 2 is a primitive root modulo 11.
   (b) Calculate a table of indices of with respect to the primitive root 2 (modulo 11).
   (c) Find all solutions to the congruence $5x^5 \equiv 6 \pmod{11}$.
3. Find the infinite continued fraction representation of the number $\frac{1}{3}(\sqrt{7} - 2)$. *Hint:* The resulting representation has the form $[a_0; \overline{a_1, a_2, a_3, a_4}]$.
4. Let p be an odd prime and $\gcd(a, p) = \gcd(b, p) = 1$. Prove that either all three of the congruences
   $$x^2 \equiv a \pmod{p}, \quad x^2 \equiv b \pmod{p}, \qquad x^2 \equiv ab \pmod{p},$$
   have a solution or exactly one of them has a solution.
5. Let $p > 3$ be a prime. Prove that
   $$\left(\frac{-3}{p}\right) = \begin{cases} 1 & p \equiv 1 \pmod{6} \\ -1 & p \equiv 5 \pmod{6}. \end{cases}$$
6. Prove that the integer $n \geq 1$ has as many representations as the sum of two squares as does the integer $2n$. That is, show that the sets
   $$\{(a, b) \in \mathbb{Z}^2 : n = a^2 + b^2\} \quad \text{and} \quad \{(u, v) \in \mathbb{Z}^2 : 2n = u^2 + v^2\}$$
   are in bijection. *Hint:* Recall the identity $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$.